36 Agricultural strategies, settlement sustainability and water dynamics in the ancient Indus Valley

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The Indus Valley Civilization (IVC) thrived around five thousand years ago during the Bronze Age in the regions of modern Pakistan and northwest India, Afghanistan and Iran. While many aspects of its social and economic organization are still debated, recent archeobotanical research has helped define the agricultural profile of the IVC, which generally consisted of a rotating system of winter (rabi) and summer (kharif) crops. However, this data has mostly been obtained from isolated case studies, and the relationship between major and smaller settlements, farming dynamics, networks and sustainability are still poorly understood.

This work, as part of the Modelling the Agricultural Origins and Urbanism in South Asia project (ModAgrO), strives to close that gap and have a better grasp of the environmental factors that might have defined the agricultural strategies in different parts of the Indus Valley, especially in the lower Indus area (modern Sindh province). For that, a diverse set of data was necessary: from hydrological dynamics to paleoclimatic and topographic data, both from the whole region and three sites of interest that were used as proxies: Bhando Qubo, in the middle of the floodplain, Taloor Jee Bhit, at the edge of the Tar desert to the east, and Mugli in a valley of the Kirthar range to the west, at the bank of a seasonal river feeding the Indus alluvial plain.

All this data has been obtained from open satellite imaging repositories (Sentinel-2 MSI, Copernicus, HydroSHEDS) through Google Earth Engine or pastclim in R for the paleoclimatical information. QGIS, a free, open-source program, has also been used to process and coalesce all data into a single geographic frame of interest. It allowed us to generate various climatic indexes like NDWI, NDVI and TWI as well as a crop suitability model through the use of FLOSS software and data. By contrasting these results with the bioarcheological data obtained from previous excavations, we explore land use and whether changes in water availability fostered distinct agricultural strategies and influenced the sustainability of local settlements.

The use of open sourced data and software ensures not only the transparency of the results but also their reproductivity and adaptability of the methodology in other studies.